Advanced Biotechnology

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CEBT II - Michaelmas 2025

UNIVERSITY OF CAMBRIDGE

Lecture 6

DNA nanotechnology

UNIVERSITY OF CAMBRIDGE

Lecture outline

1) Not merely the secret of life: DNA as a material
2) Structural DNA nanotechnology
3) DNA origami
4) Dynamic DNA nanotechnology

Learning outcomes

1. Describe what makes DNA an excellent material for nanoscale engineering
2. Describe the principles and pipelines of structural DNA nanotechnology
3. Describe the strategy behind DNA origami
4. Describe principles of dynamic DNA nanotechnology
5. Quantitatively describe toehold mediated strand displacement

Lecture outline

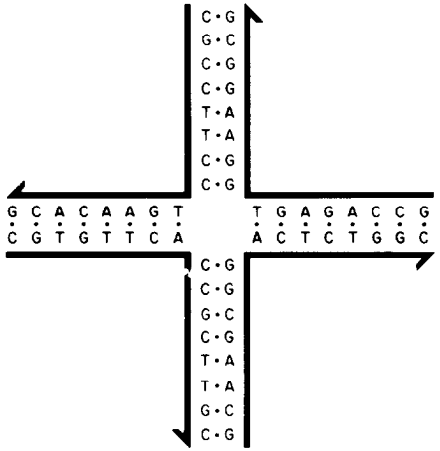
1) **“Not merely the secret of life”: DNA as a material**
2) Structural DNA nanotechnology
3) DNA origami
4) Dynamic DNA nanotechnology

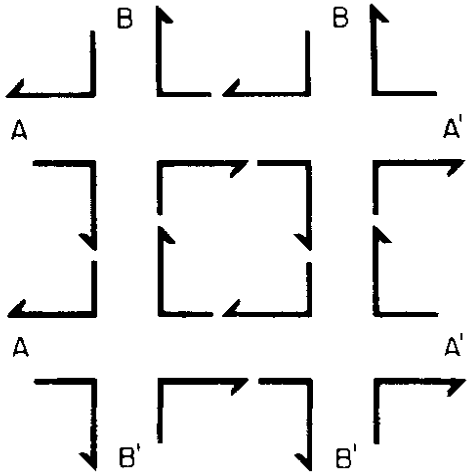
DNA nanotechnology, building with DNA bricks

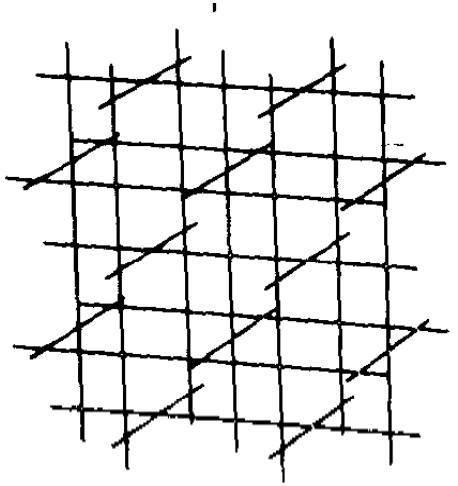
Ned Seeman 1945-2021

"Nucleic acid junctions and lattices", *Journal of Theoretical Biology*, 99(2), 237-247, 1982

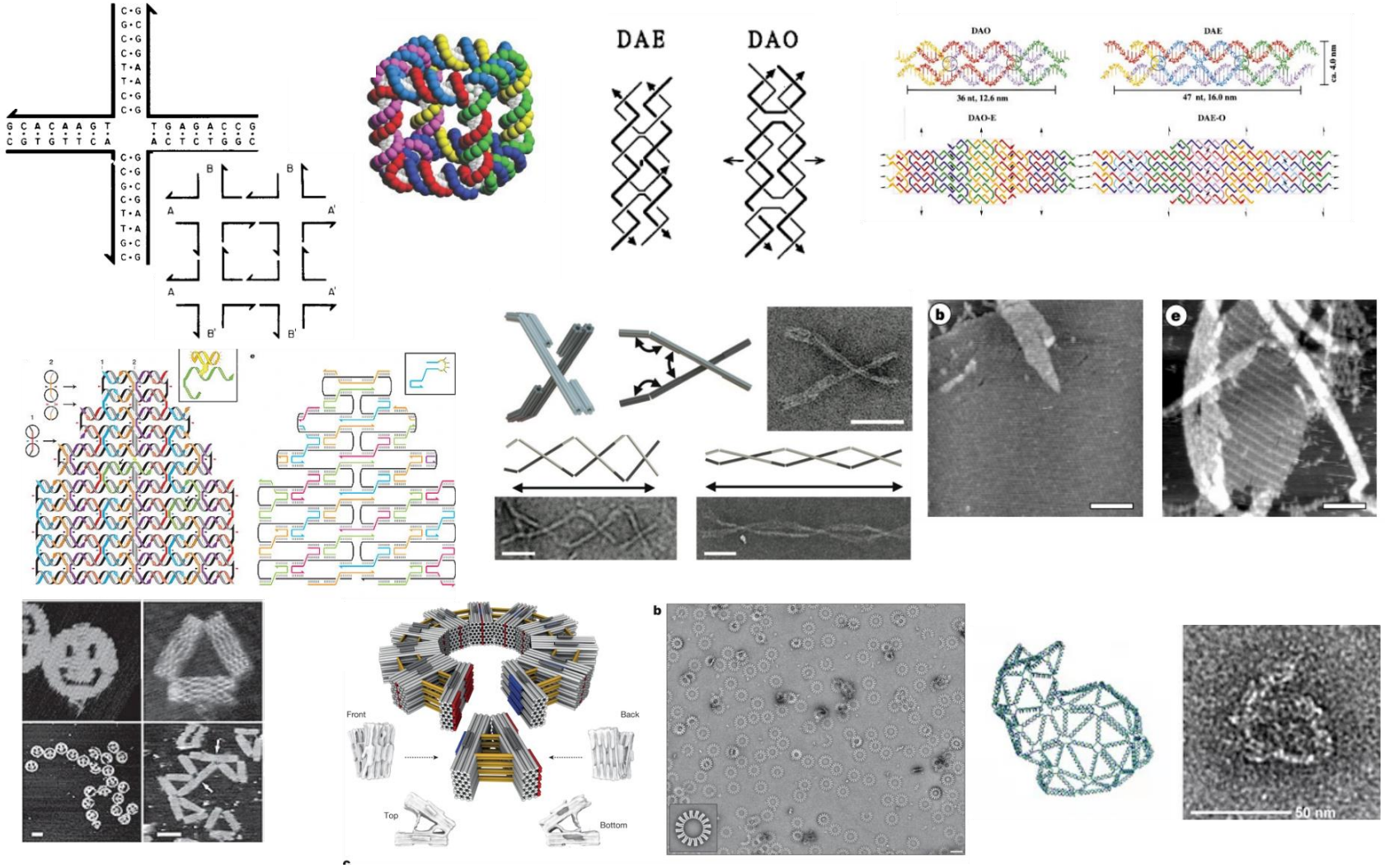
"It appears to be possible to generate covalently joined three-dimensional networks of nucleic acids which are periodic in connectivity and perhaps in space."







Building with DNA bricks



Seeman and Sleiman Nat. Rev. Mater., 17068, 2018

DNA nanotechnology

- **DNA nanotechnology** is a collection of **tools and methods** (experimental, theoretical, computational) that exploits the physical and chemical properties of DNA (and RNA) to **control structure and dynamics at the nanoscale**

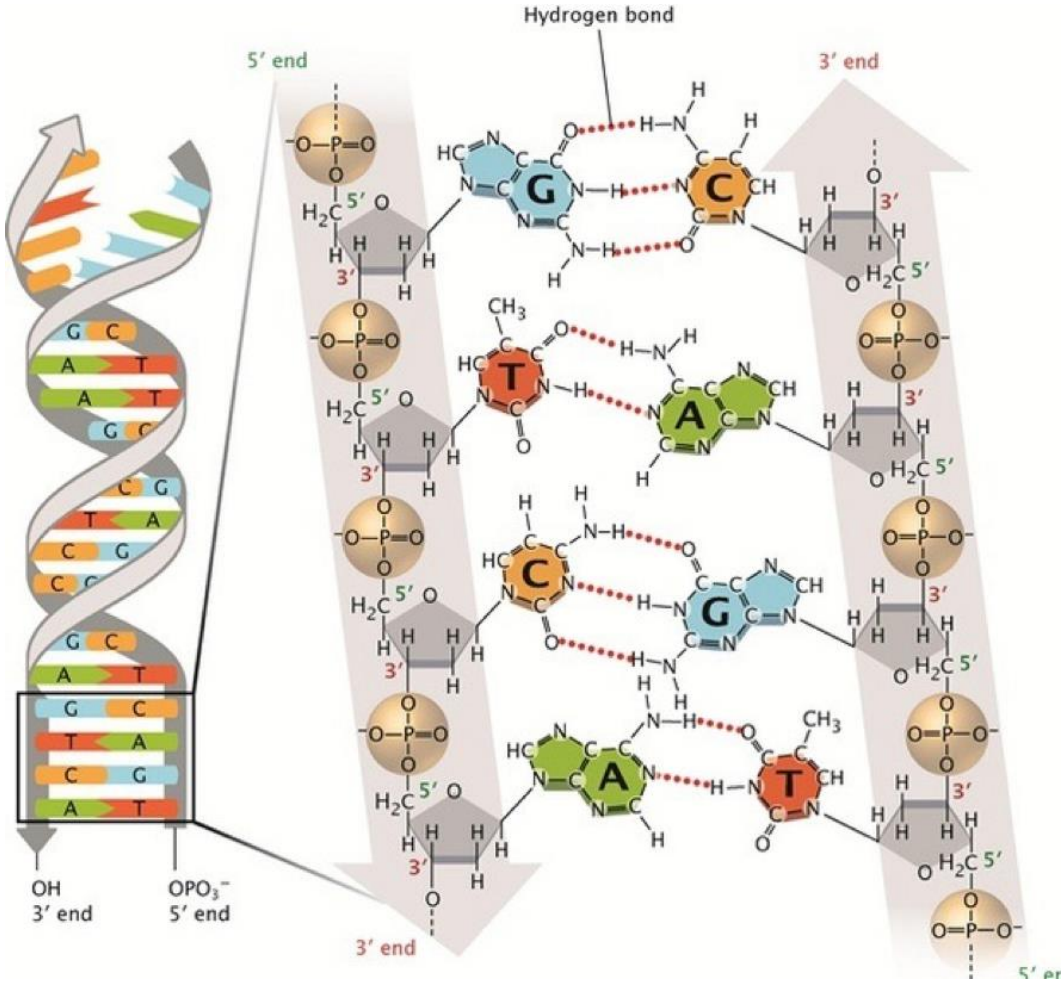
- Applications to healthcare, sensing, materials, microscopy, synthetic biology...

- Traditionally, it is divided in **structural** and **dynamic** DNA nanotechnology

“Not merely the secret of life”

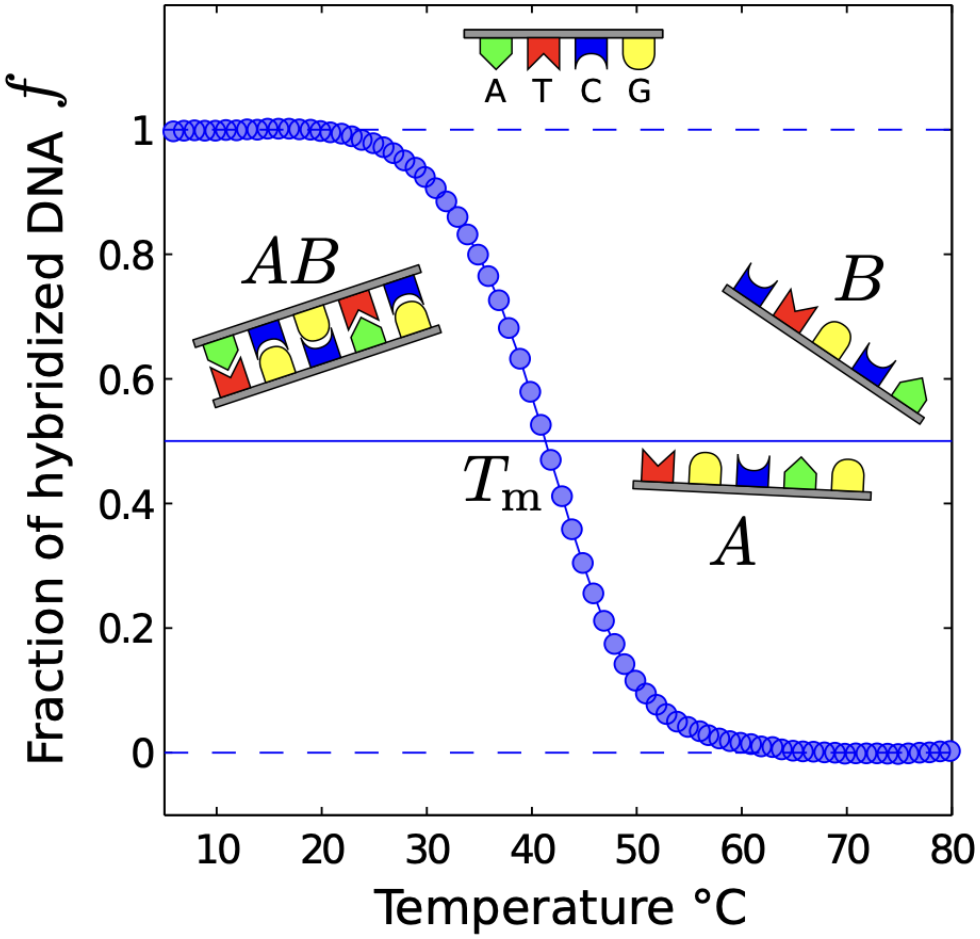
We learned (extensively) about the biological role of DNA, but DNA has many interesting properties as a material

- Chemically stable and not easily degraded (different from RNA)
- DNA oligos with arbitrary sequence can be chemically synthesised, at decreasing price
- Synthetic DNA can be easily chemically modified (e.g. to append fluorophores, quenchers, linkers for conjugation)



“Not merely the secret of life”

- Interesting polymer physics properties (single stranded DNA is much more flexible than double-stranded DNA)
- Nucleic acid thermodynamics is very well understood and predictable (recall simple models for computing ΔG^0, $T_m \ldots$)
- DNA hybridisation is thermally reversible under near-environmental conditions



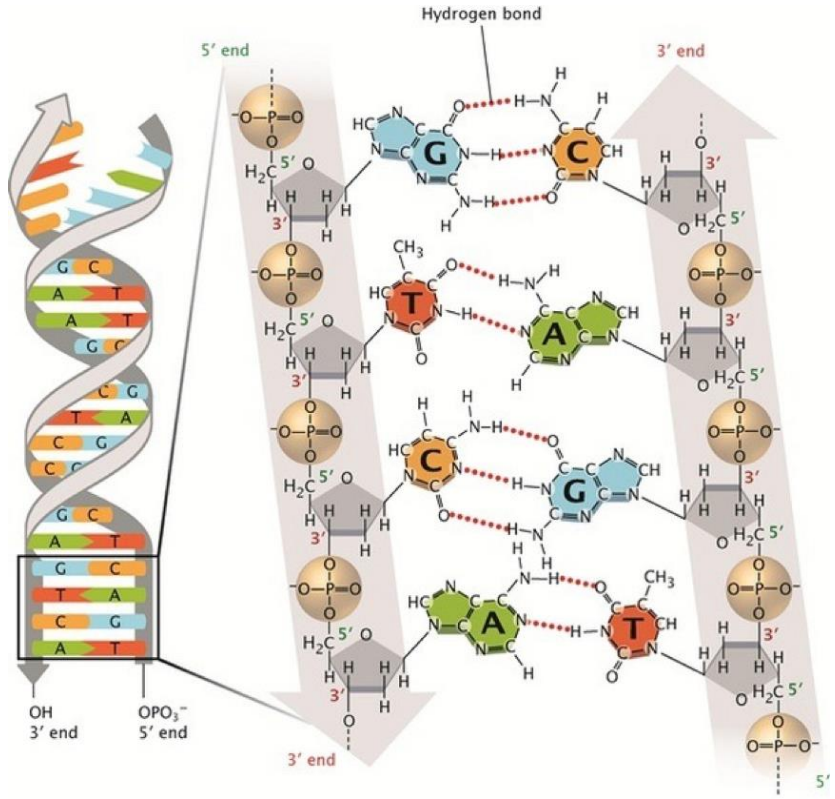
Base pairing selectivity

- **Base pairing interactions are exquisitely selective**
- Consider two complementary 12 nt strands at equal concentration of $1\ \mu\mathrm{M}$ and $T = 293.15\ \mathrm{K}$. Hybridisation free energy and melting temperature are (for the specific sequence I chose): $\Delta G = -16.8\ \mathrm{kcal\ mol^{-1}}$, $T_m = 53.2°\mathrm{C}$
- Now consider two random 12 nt strands in the same conditions. For these, I calculated: $\Delta G = -2.9\ \mathrm{kcal\ mol^{-1}}$, $T_m = -30.7°\mathrm{C}$
- Note the **massive difference** in hybridisation free energy and melting temperature. Complementary strands need to me heated up to over $50°\mathrm{C}$ for melting, non-complementary strands would not hybridise unless you cool them way below room temperature

Sequence orthogonality

- For a DNA N-mer the number of possible sequences is 4^N
- For a 12-mer, this is 1.7×10^7 different sequences
- Of these most will barely interact with each other (high ΔG, low T_m), while only a small sub-set would be somewhat complementary
- **Possibility of designing a very large number of molecules with orthogonal interactions**!

A material with many desirable properties

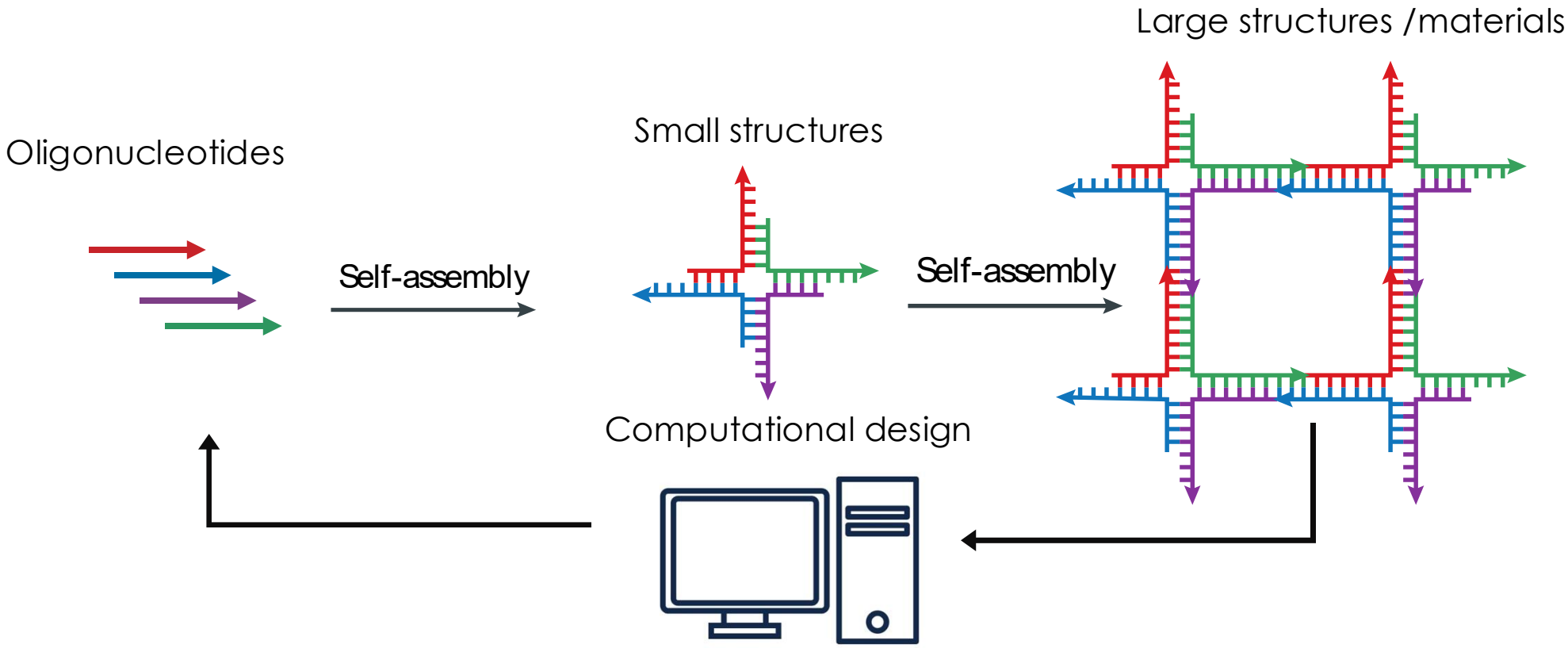


- **Bond selectivity and reversibility**
- **Well understood and programmable thermodynamics and kinetics**
- Built-in stimuli-responsiveness
- Biocompatible
- Robust against degradation
- Steadily decreasing synthesis price
- Easy functionalization

Lecture outline

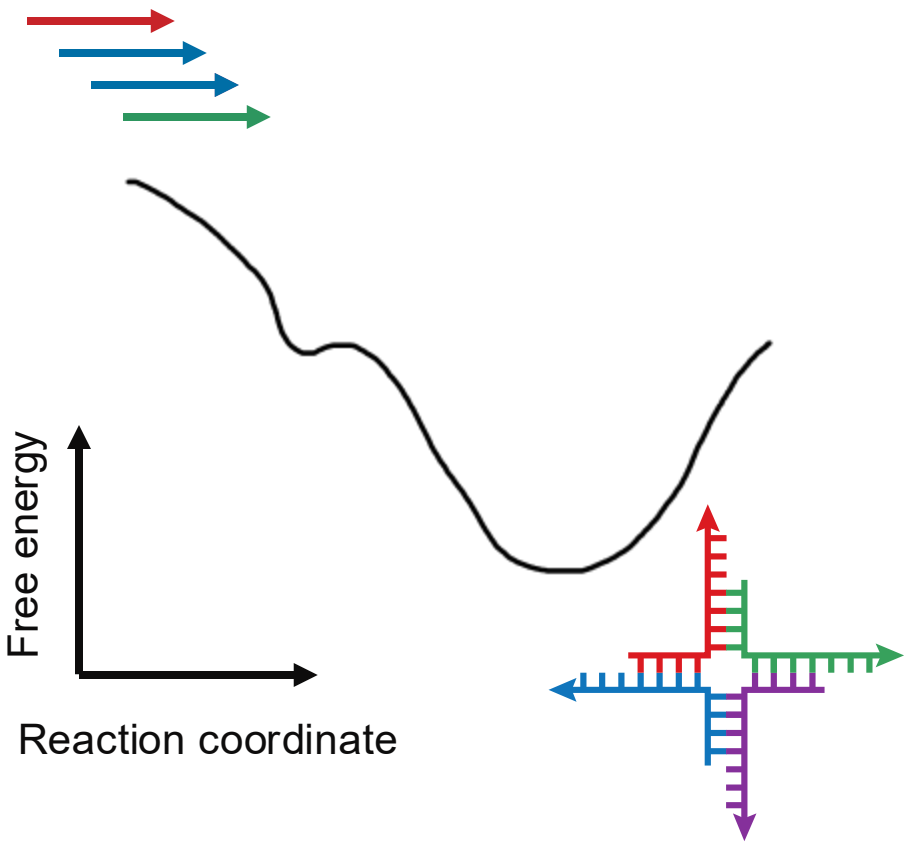
1) "Not merely the secret of life": DNA as a material
2) **Structural DNA nanotechnology**
3) DNA origami
4) Dynamic DNA nanotechnology

Oligonucleotide self-assembly



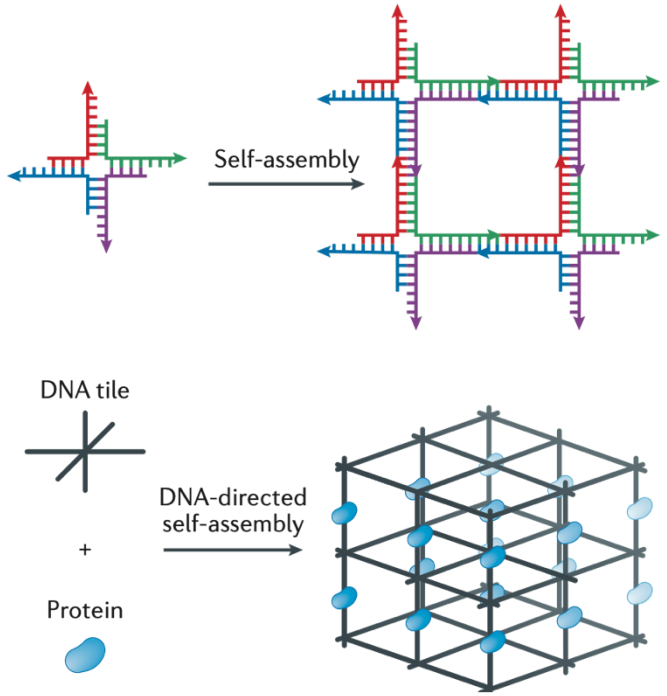
- We start from a given nanostructure or material we want to build with DNA
- Then use computational design to figure out a set of oligonucleotides that would "self-assemble" into the target structure
- Computational tools use knowledge of DNA thermodynamics and leverage base-pairing selectivity

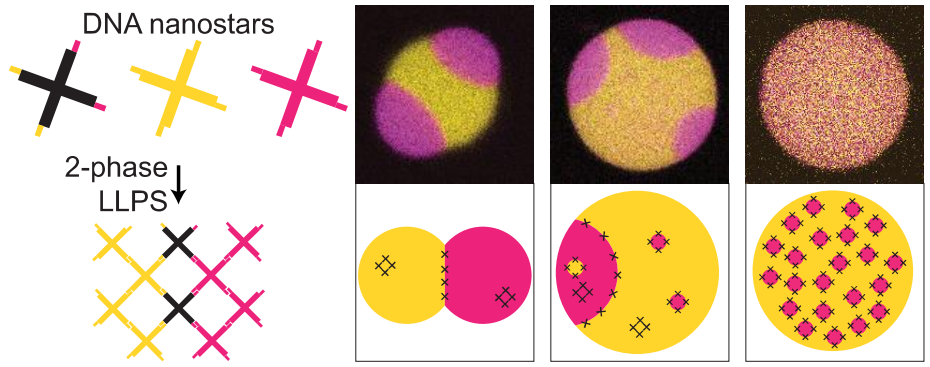
Equilibrium self-assembly



- Target structure minimizes free energy, which is dominated by base pairing, hence predictable
- Slow thermal annealing from above T_m (typically to room temperature) ensures equilibration and avoids kinetic (metastable) traps

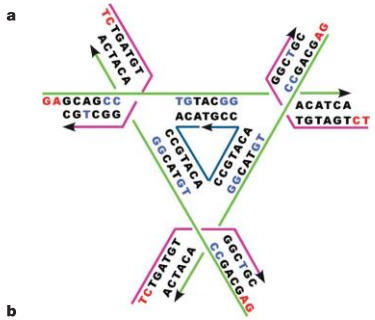
DNA junctions

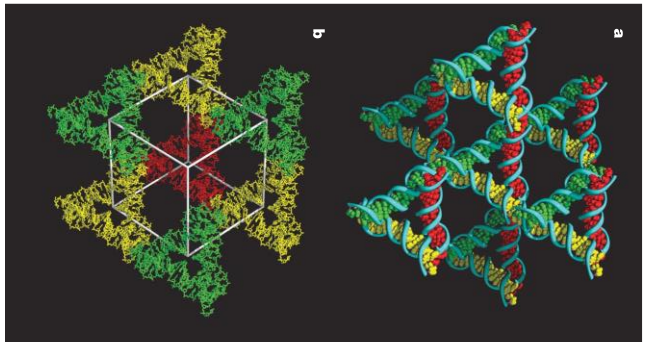




- Use DNA junctions that bind to each other with ssDNA "sticky ends" to build crystalline networks
- Possibly useful for aiding crystallization of proteins or generally arrange molecules in 3D
- Simple junctions like these **do not crystallize** due to high flexibility. They for hydrogels and liquid materials instead

Tensegrity triangles

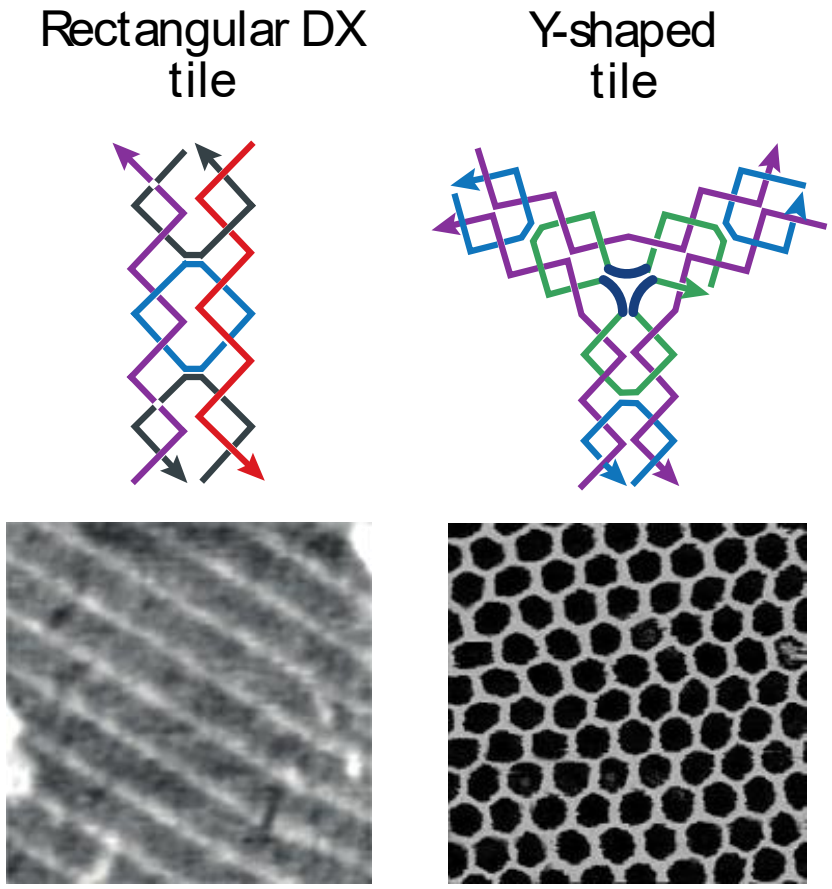




J. Zheng et al., Nature 461 2009

- Ned Seeman eventually succeeded in crystalizing DNA nanostructures in 3D. It took 27 years!
- Achieved with small, **rigid structures** dubbed **tensegrity triangles**

Not just junctions



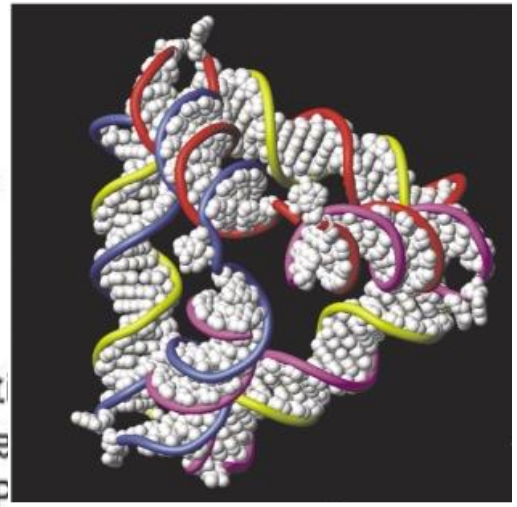
Winfree et al. Nature 394, 539–544 (1998)
Seeman and Sleiman Nat. Rev. Mater., 17068, 2018

- Several two-dimensional "tiles" have been assembled from DNA oligoncleotudes.
- **Milestone: Double crossover (DX) tile**
- Able to sustain 2D crystallization (easier than 3D, note dates)
- Can be used to regularly arrange functional elements on a surface (e.g. nanoparticles)

3D nanostructures

823 (1990).
10 (1990).
. **9,** 3501–3517 (1989).
Acids Res. **18,** 5433–5441 (1990).
57–881 (1982).
V. Nature **321,** 133–139 (1986).

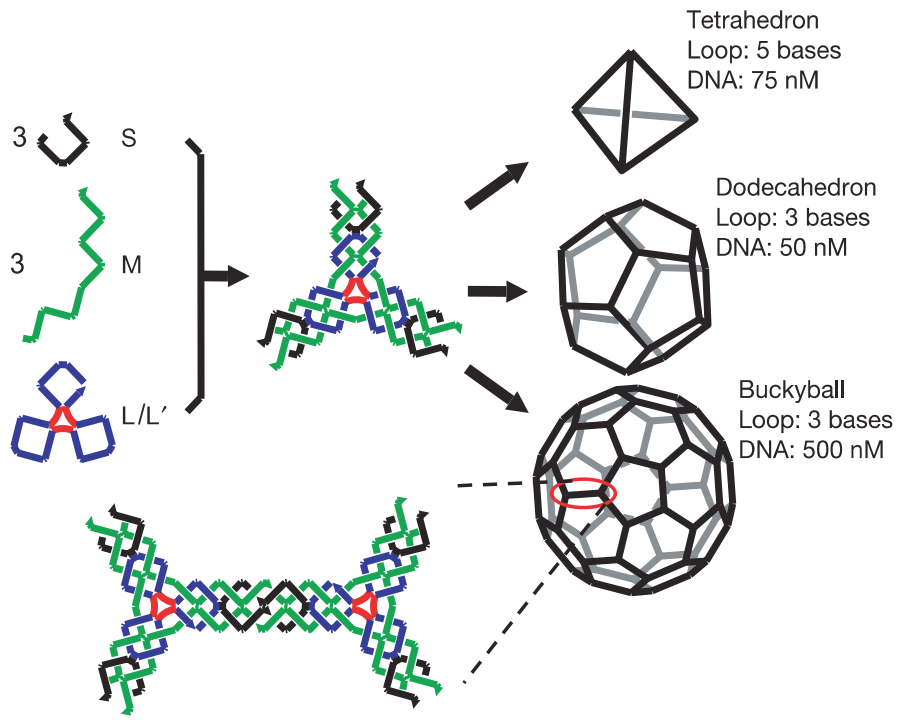
Ritter for synthesis of oligonucleot
scussions. This work was supporte
T.R.C. is American Cancer Society F
hank the W. H. Keck Foundation for



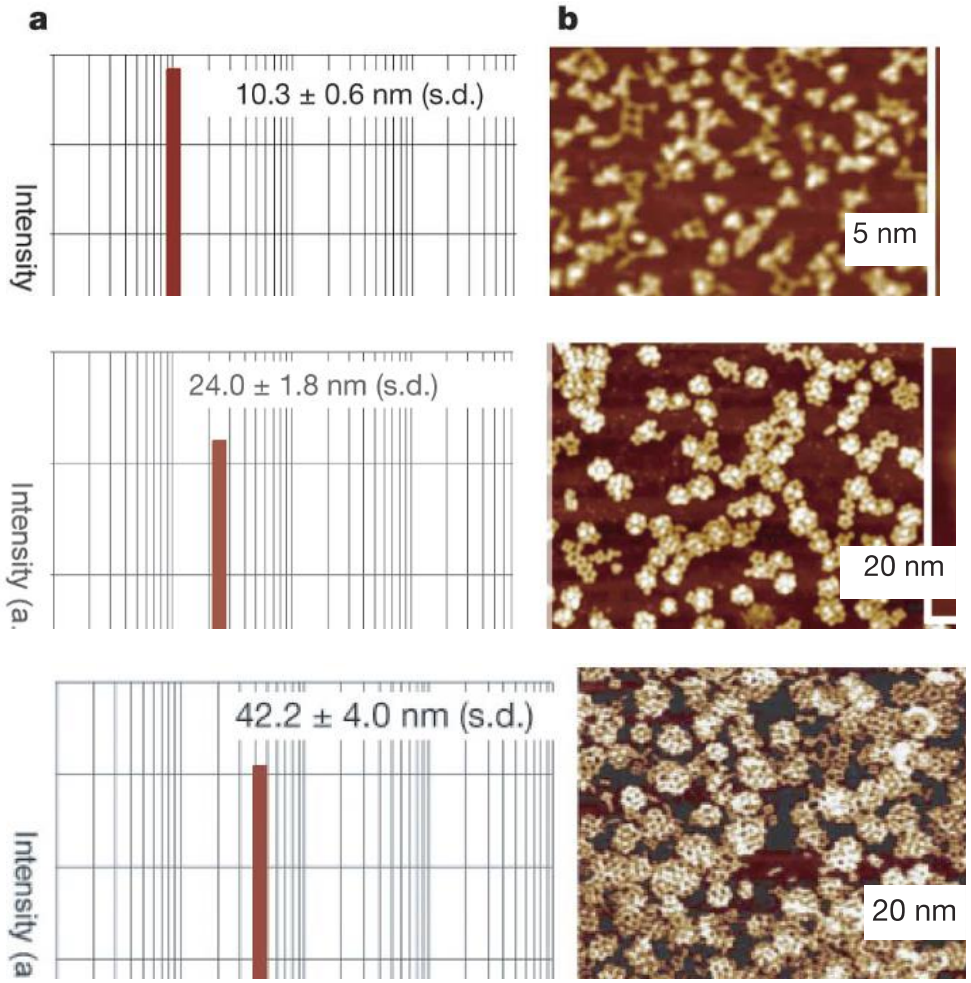
Left: Chen and Seeman, Nature, 350, 631, 1991
Right: Chem Comm, 1372, 2004

- DNA cube, 1st 3D nanostructure (also by Seeman and co). Not really 3D cube, as cube is floppy!
- 1st true 3D motif was a tetrahedron by Turberfield and co

Polyhedra from crossover tiles



He et al. Nature 452, 198 2008



Lecture outline

1) "Not merely the secret of life": DNA as a material
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4) Dynamic DNA nanotechnology

Paul Rothemund's DNA origami

Vol 440|16 March 2006|doi:10.1038/nature04586 nature

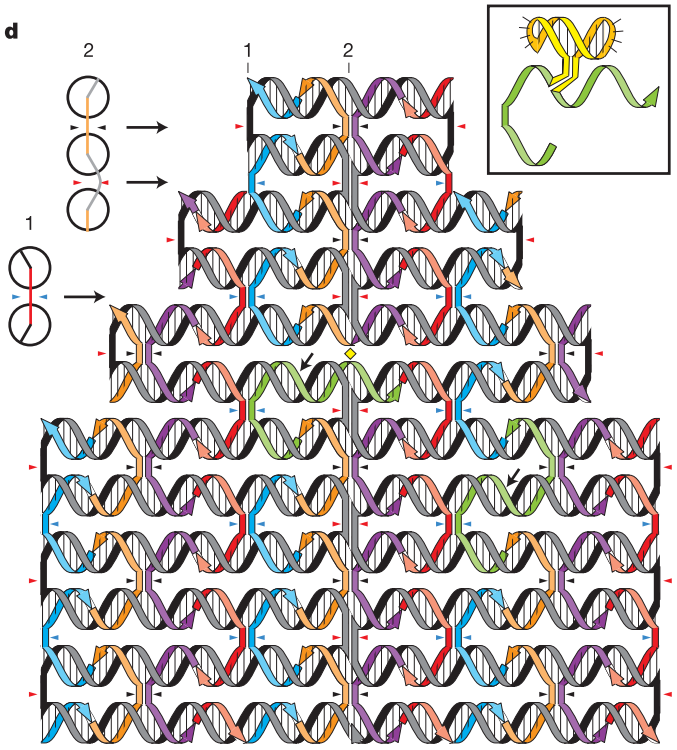
ARTICLES

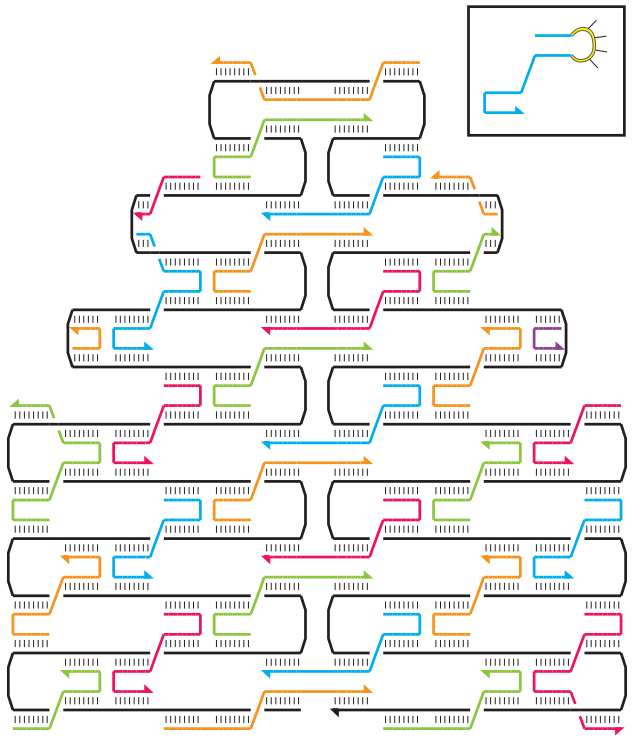
Folding DNA to create nanoscale shapes and patterns

Paul W. K. Rothemund[1]

'Bottom-up fabrication', which exploits the intrinsic properties of atoms and molecules to direct their self-organization, is widely used to make relatively simple nanostructures. A key goal for this approach is to create nanostructures of high complexity, matching that routinely achieved by 'top-down' methods. The self-assembly of DNA molecules provides an attractive route towards this goal. Here I describe a simple method for folding long, single-stranded DNA molecules into arbitrary two-dimensional shapes. The design for a desired shape is made by raster-filling the shape with a 7-kilobase single-stranded scaffold and by choosing over 200 short oligonucleotide 'staple strands' to hold the scaffold in place. Once synthesized and mixed, the staple and scaffold strands self-assemble in a single step. The resulting DNA structures are roughly 100 nm in diameter and approximate desired shapes such as squares, disks and five-pointed stars with a spatial resolution of 6 nm. Because each oligonucleotide can serve as a 6-nm pixel, the structures can be programmed to bear complex patterns such as words and images on their surfaces. Finally, individual DNA structures can be programmed to form larger assemblies, including extended periodic lattices and a hexamer of triangles (which constitutes a 30-megadalton molecular complex).

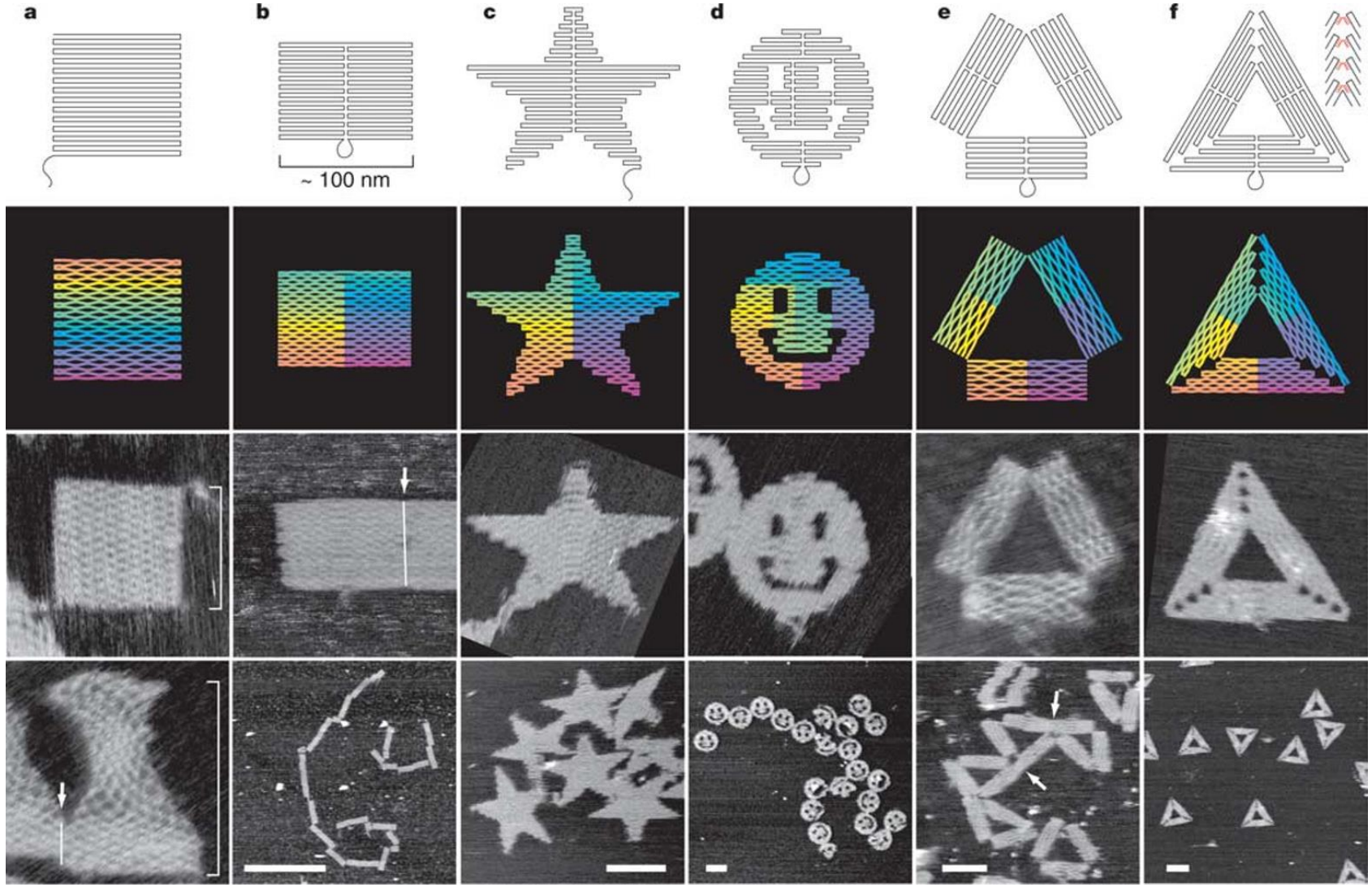
Paul Rothemund's DNA origami



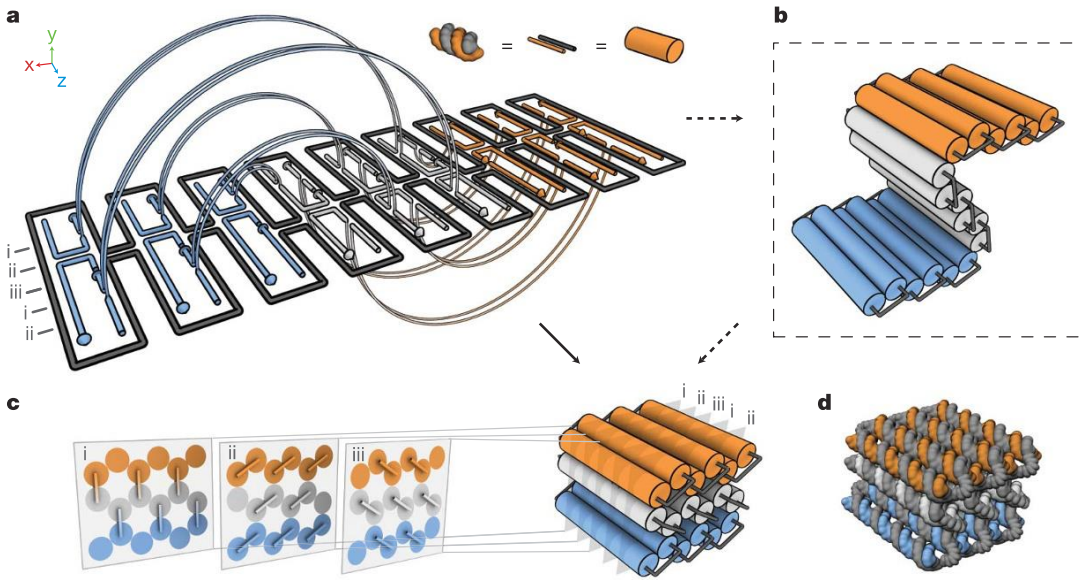


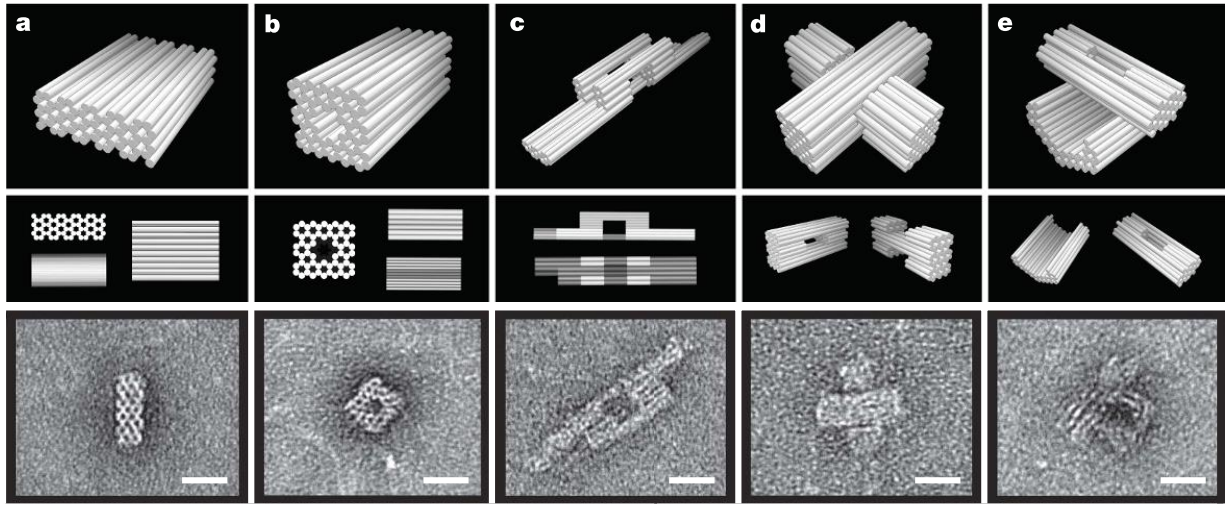
- ~7 knt ssDNA scaffold from ssDNA virus (M13 bacteriophage)
- "Folded" using hundreds of staple strands
- Crossovers link together adjacent helices
- Initially 2D only

Paul Rothemund's DNA origami



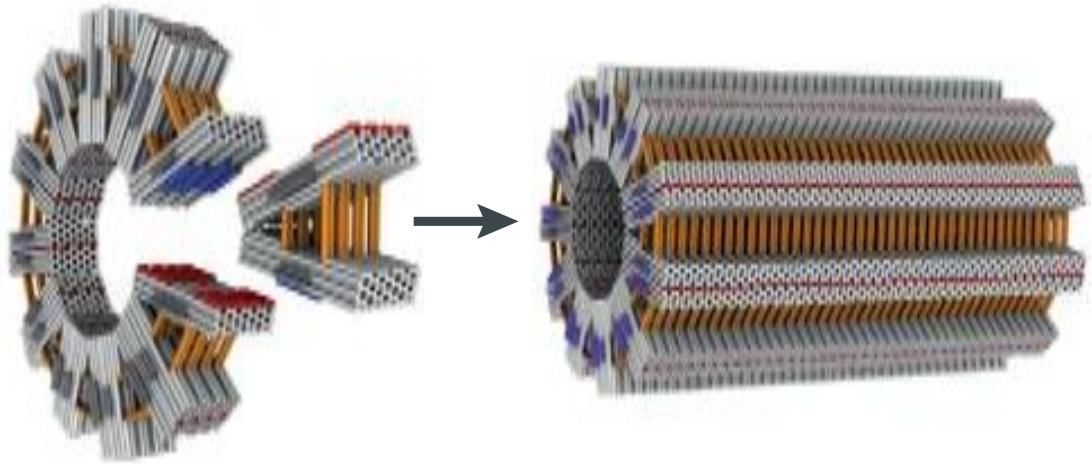
3D DNA origami



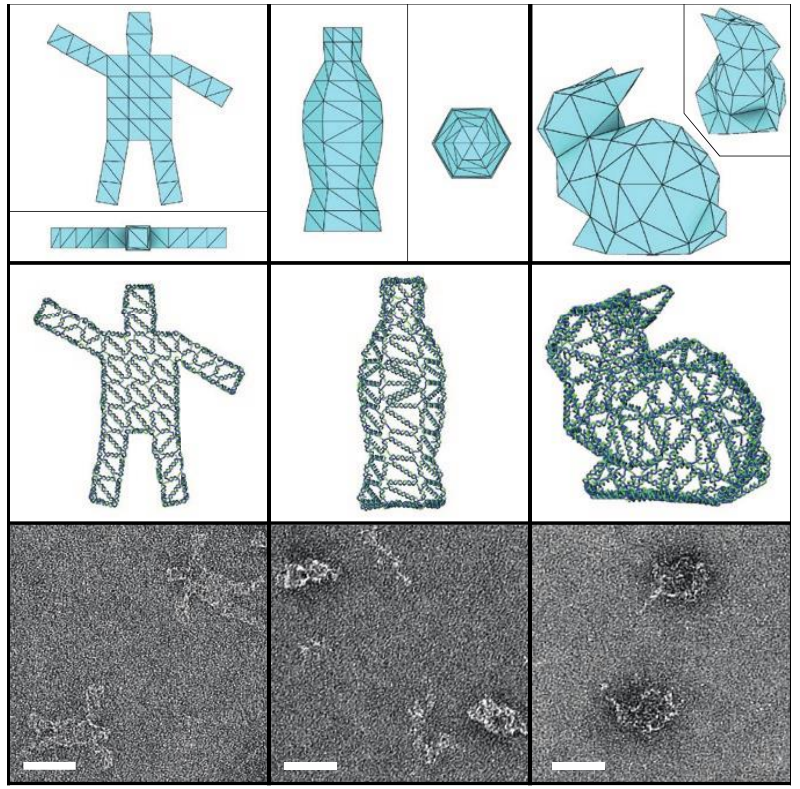


Douglas et al. Nature 459, 414 2009

Origami polymerization and wireframe designs

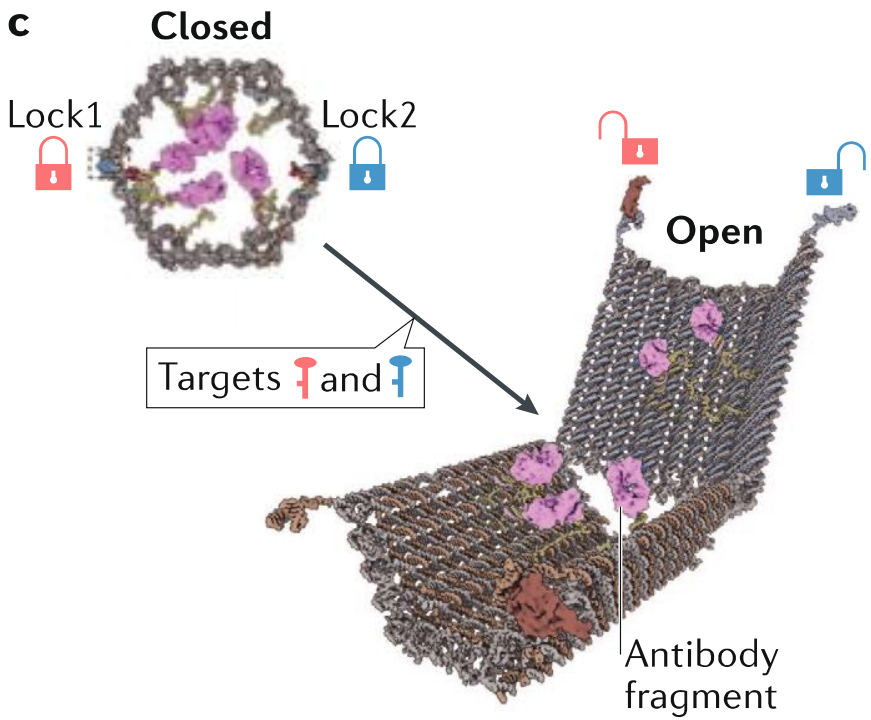


Gigadalton-scale assembly of origami
Wagenbauer et al Nature 552, 78–83 2017

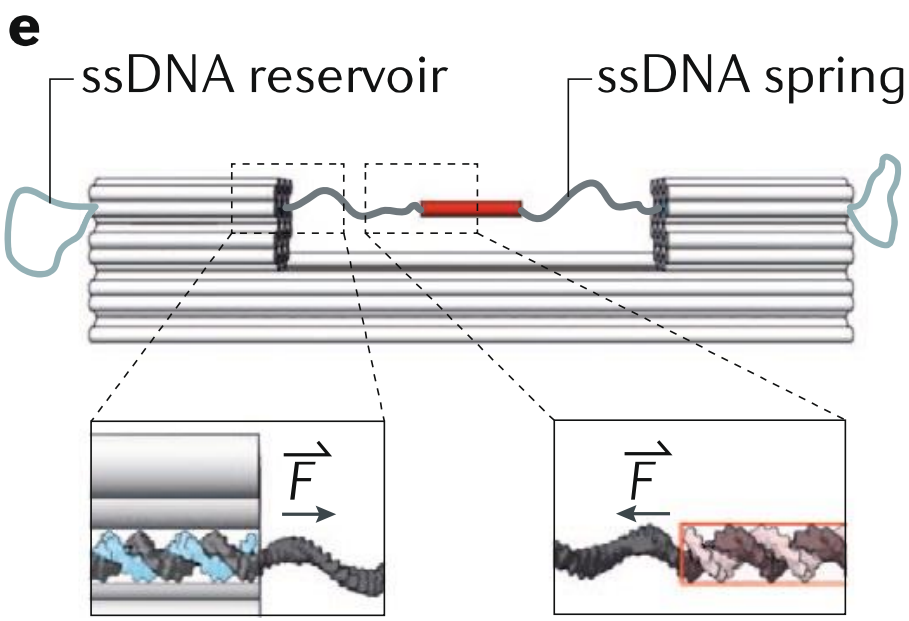


Automated design of wireframe DNA origami
Benson et al. Nature volume 523, 441–444 2015

Applications…

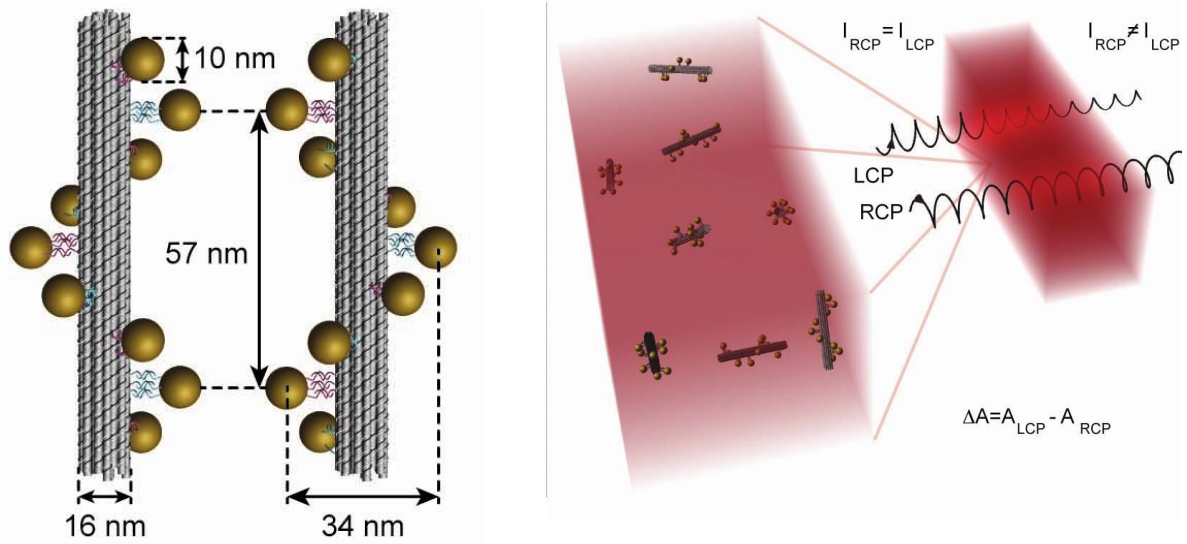


Drug delivery
Douglas et al Science, 335(6070) 831-834 2012



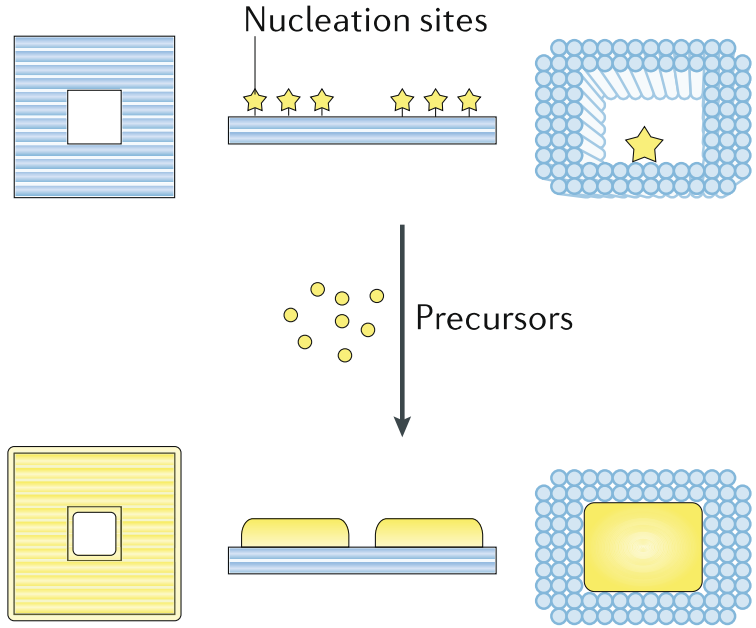
Biophysics: measuring molecular forces
Science, 354, 6310, 305-307, 2016

Applications…



Optical materials. Filtering for circularly polarised light

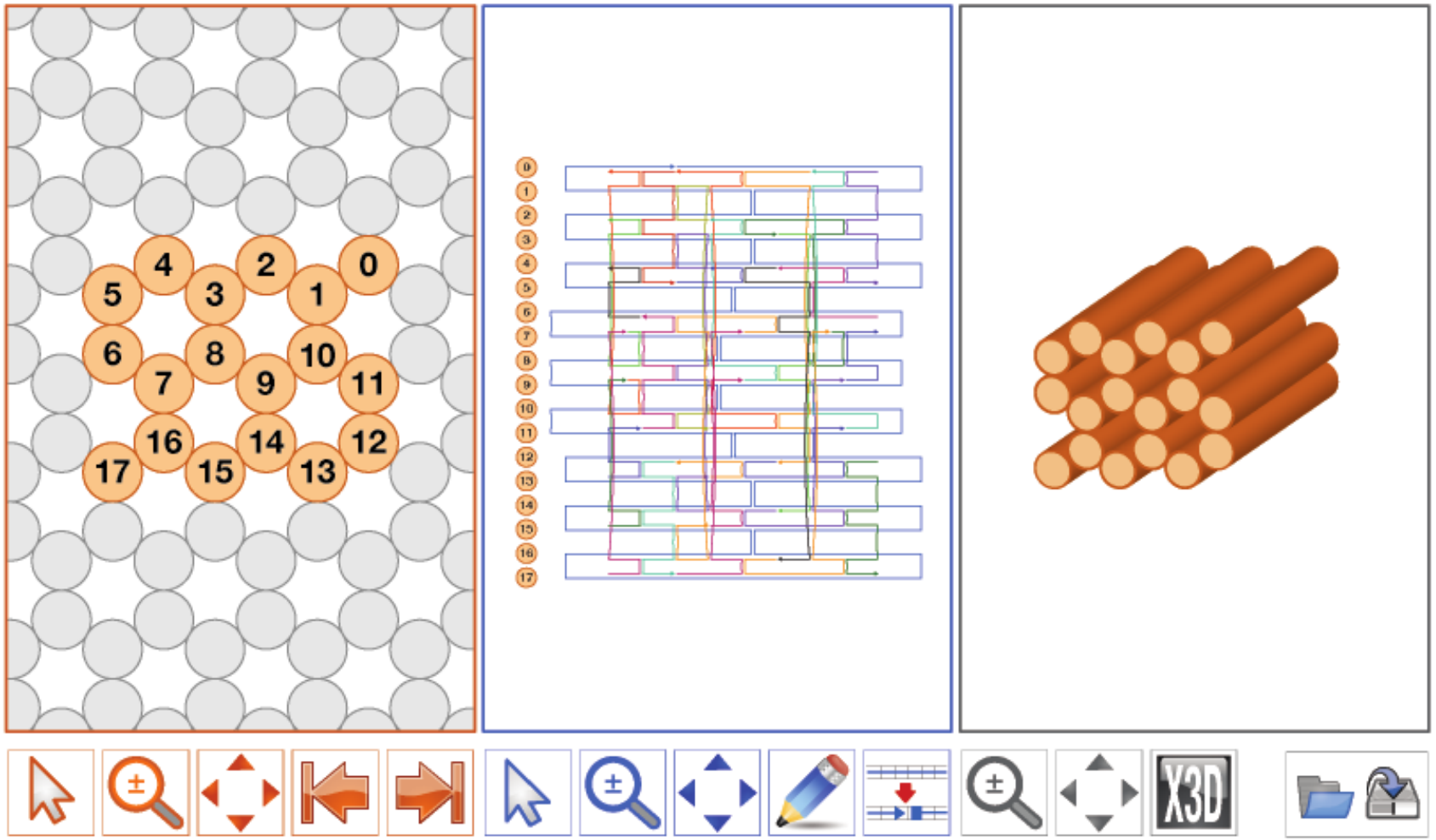
Kuzyk, A. et al. Nature 483, 311–314 2012.



Templated growth of materials

Science, 346, 6210, 717, 2014

Origami design: caDNAno



Douglas et al, Nucleic Acids Research, 37, 15, 2009

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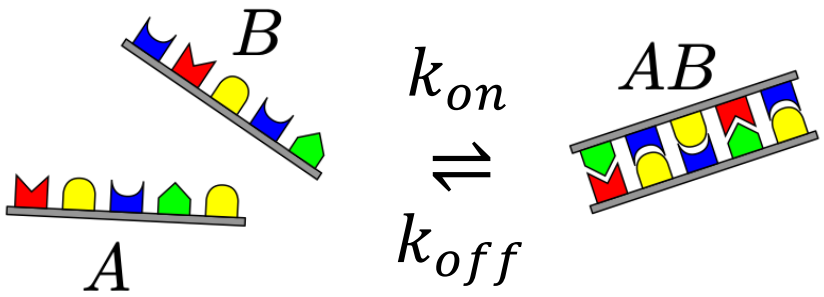
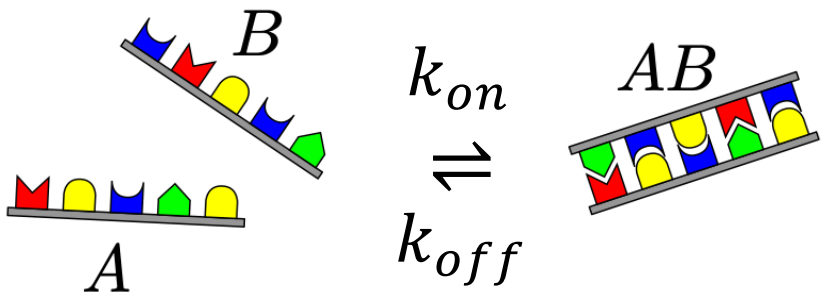
Dynamic DNA nanotechnology



- Building DNA nanostructures that can change shape, bind/unbind, reconfigure etc. in response to stimuli
- Stimuli include presence of other molecules, changes in environmental conditions…
- Applications to sensing, diagnostics, molecular computing…

Controlling kinetics of DNA hybridization

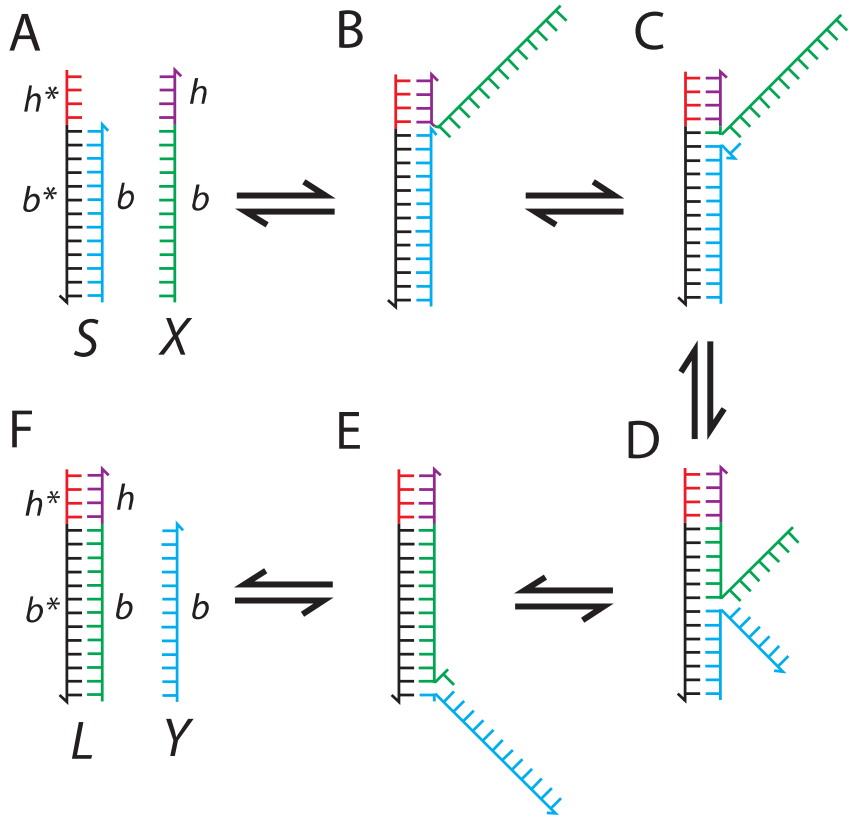


$$k_{on} \sim 10^4 - 10^6 \ \mathrm{s^{-1}M^{-1}}$$

$$k_{off} = k_{on}\,\rho_0 \exp\left(\frac{\Delta G^0}{RT}\right)$$

- Actuating DNA nanostructures often requires breaking and re-forming hybridization links
- Unbinding rates of oligos depend on ΔG^0, hence **kinetics of reconfiguration is linked to thermodynamic stability**
- **This is undesirable**! It would be much better to be able to control the kinetics of reconfiguration (more or less) independently on thermodynamic stability.

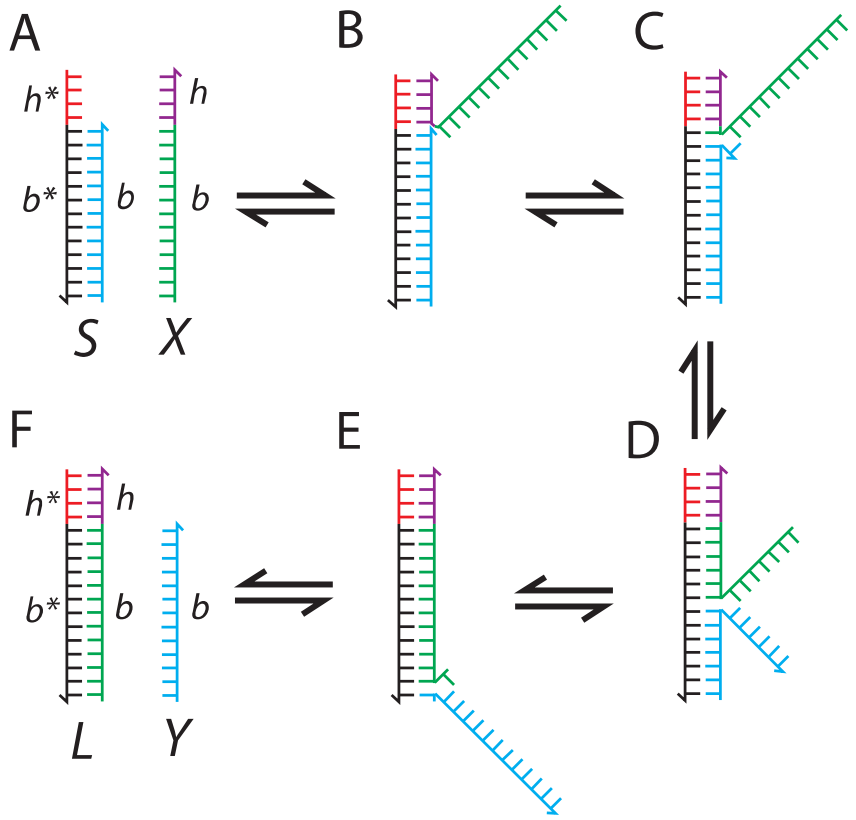
Toehold-mediated strand displacement



Simmel et al. Chem. Rev 119, 6326−6369, 2019

- The objective is breaking the duplex S and forming a new duplex L, in which strand X (invader) has replaced strand Y (incumbent)
- Thermal dissociation of duplex S would take a very long time because of the high free energy cost of unbinding several bases.
- **Toehold-mediated strand displacement** (**TMSD**) catalyzes the process of switching bonds by co-localizing X and S using a toehold (domains h and its complementary h*)

Toehold-mediated strand displacement



Simmel et al. Chem. Rev 119, 6326−6369, 2019

A) S and X are in solution

B) Domain h in X binds to h* in S

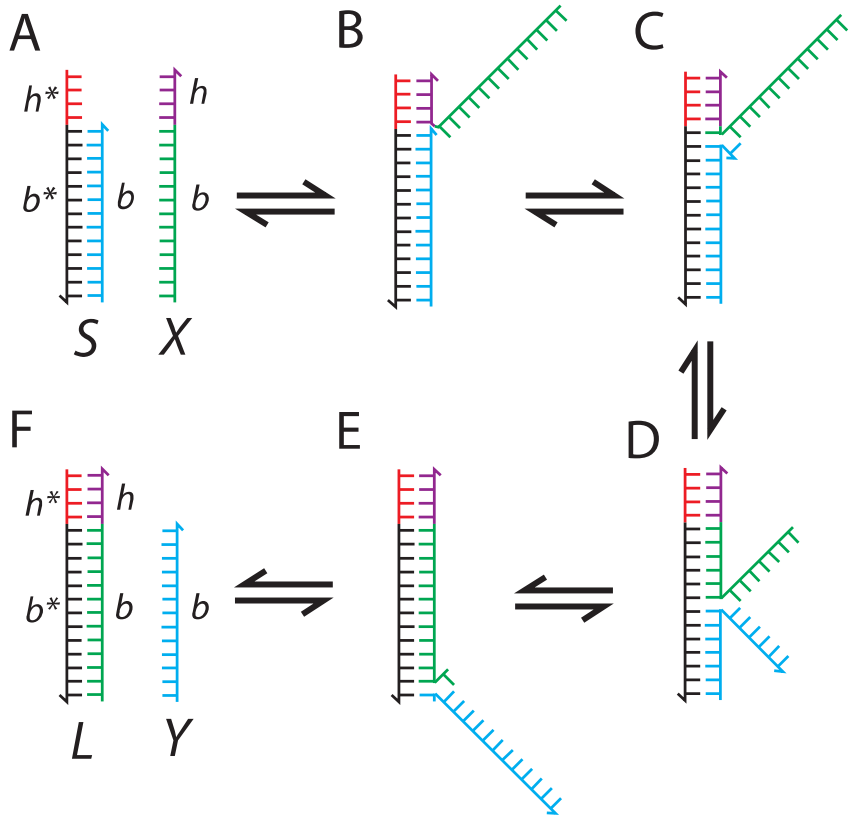
C) X starts to displace Y **one base at the time**

D) The displacement proceeds through a three-way branch migration process. This is a **diffusive process** rather than thermally activated, and can go either way

E) Eventually, branch migration proceeds in the desired direction and is completed

F) Y is released and duplex L is formed

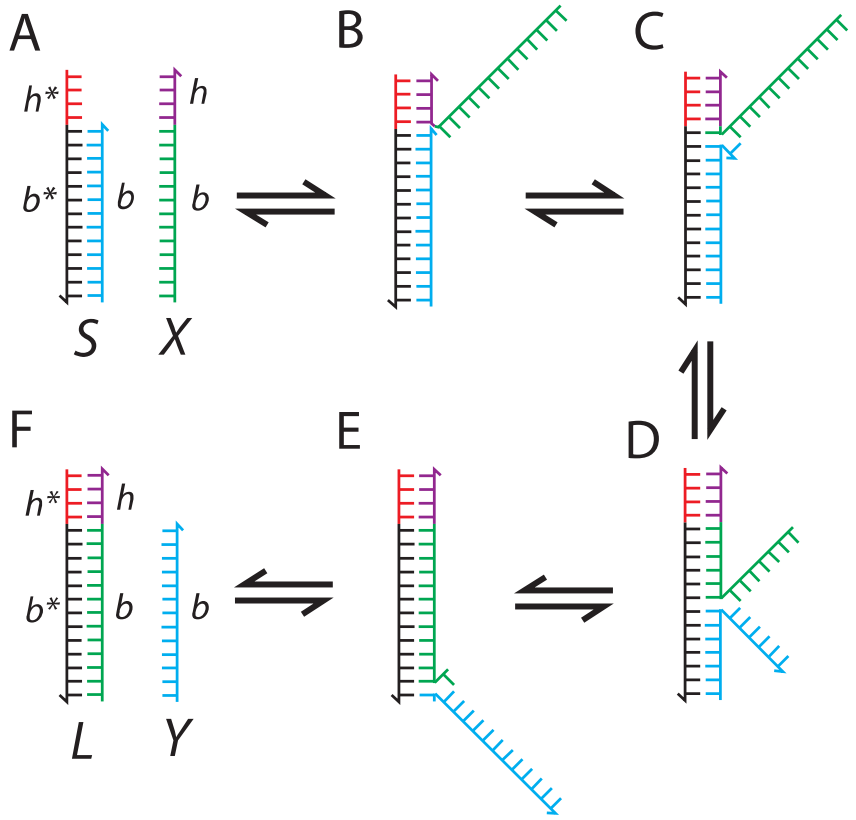
Toehold-mediated strand displacement



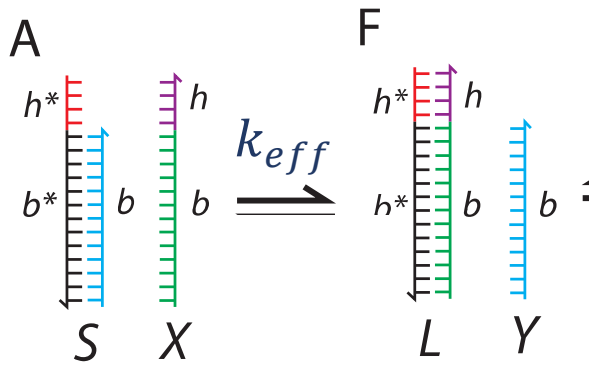
Simmel et al. Chem. Rev 119, 6326−6369, 2019

- The final step (E-F) is irreversible, because there is no toehold for the reverse reaction
- If h and h* are sufficiently long, TMSD can be much faster than duplex exchange mediated by thermal denaturation. This is because the **free energy barrier for the process is much lower**! : at no point domains b and b* detach completely
- TMSD is **thermodynamically driven**. Duplex S has fewer base-pairs than duplex L
- However, the rate of the displacement is (more or less) **independent on the absolute stability of the duplexes**, as determined by domains b and b*!

Toehold-mediated strand displacement



Simmel et al. Chem. Rev 119, 6326−6369, 2019



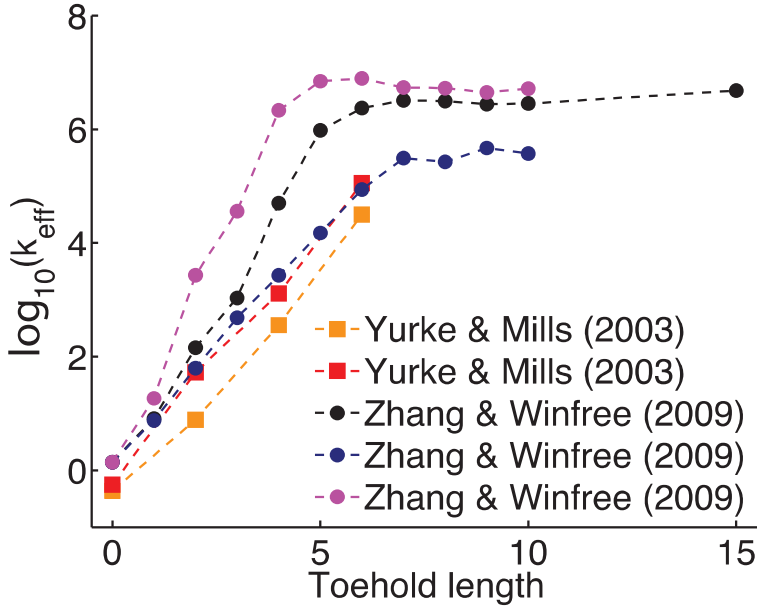
TMSD can be approximated as a second-order reaction with an effective rate constant

$$k_{eff} \approx \frac{k_{on}}{k_{off}/k_{BM} + 1}$$

Where k_{on} and k_{off} are the binding and unbinding rates of the toehold and k_{BM} is the branch migration rate

Toehold-mediated strand displacement

$$k_{eff} \approx \frac{k_{on}}{k_{off}/k_{BM} + 1}$$



Simmel et al. Chem. Rev 119, 6326−6369, 2019

- For **short toehold** $k_{off} \gg k_{BM}$, hence

$$k_{eff} \approx \frac{k_{BM} k_{on}}{k_{off}} = k_{BM}\, \rho_0 \exp\left(-\frac{\Delta G^0}{RT}\right)$$

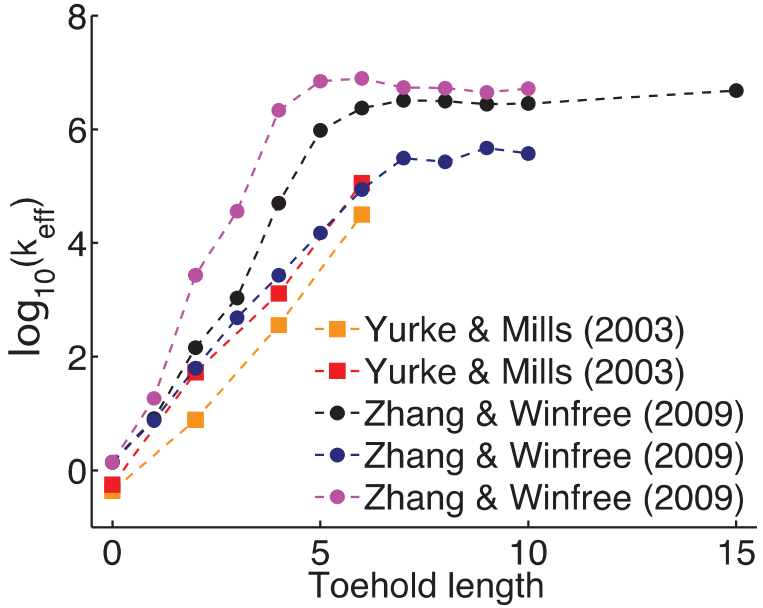
Here, k_{eff} **increases exponentially** with $-\Delta G^0$, and hence with **toehold length**

- For **long toehold** $k_{off} \ll k_{BM}$, hence $k_{eff} \approx k_{on}$. Here, k_{eff} plateaus and becomes independent on toehold length

- Crossover between the two regimes occurs for toeholds ~5-7 nt

Toehold-mediated strand displacement

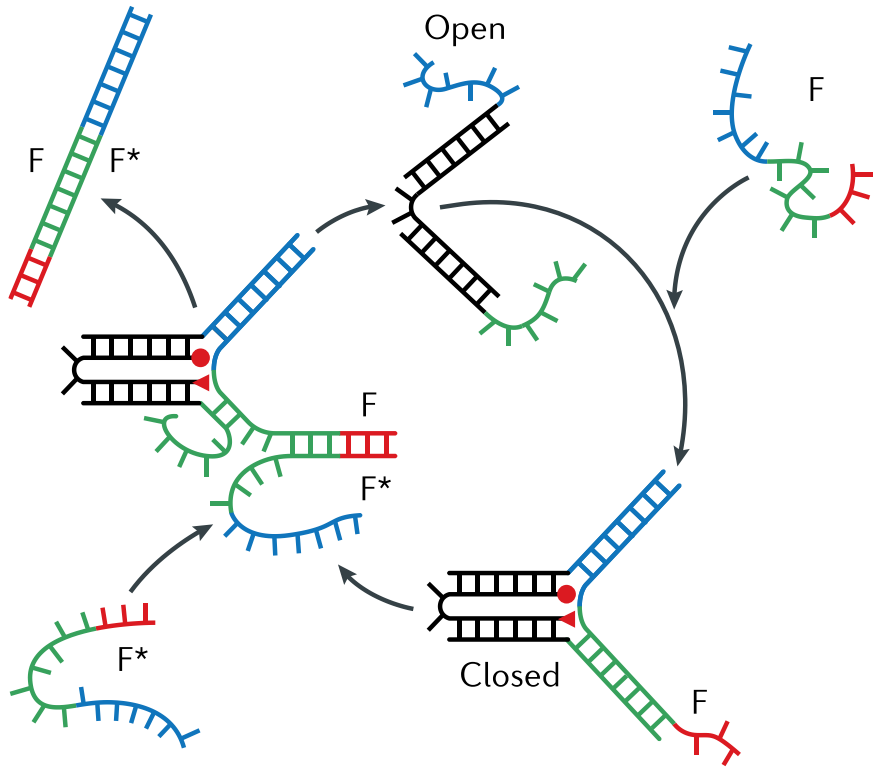
$$k_{eff} \approx \frac{k_{on}}{k_{off}/k_{BM} + 1}$$



- Crossover between the two regimes occurs for toeholds ~5-7 nt
- By changing toehold length (binding strength), **we can tune k_{eff} over ~6 orders of magnitude**

Simmel et al. Chem. Rev 119, 6326−6369, 2019

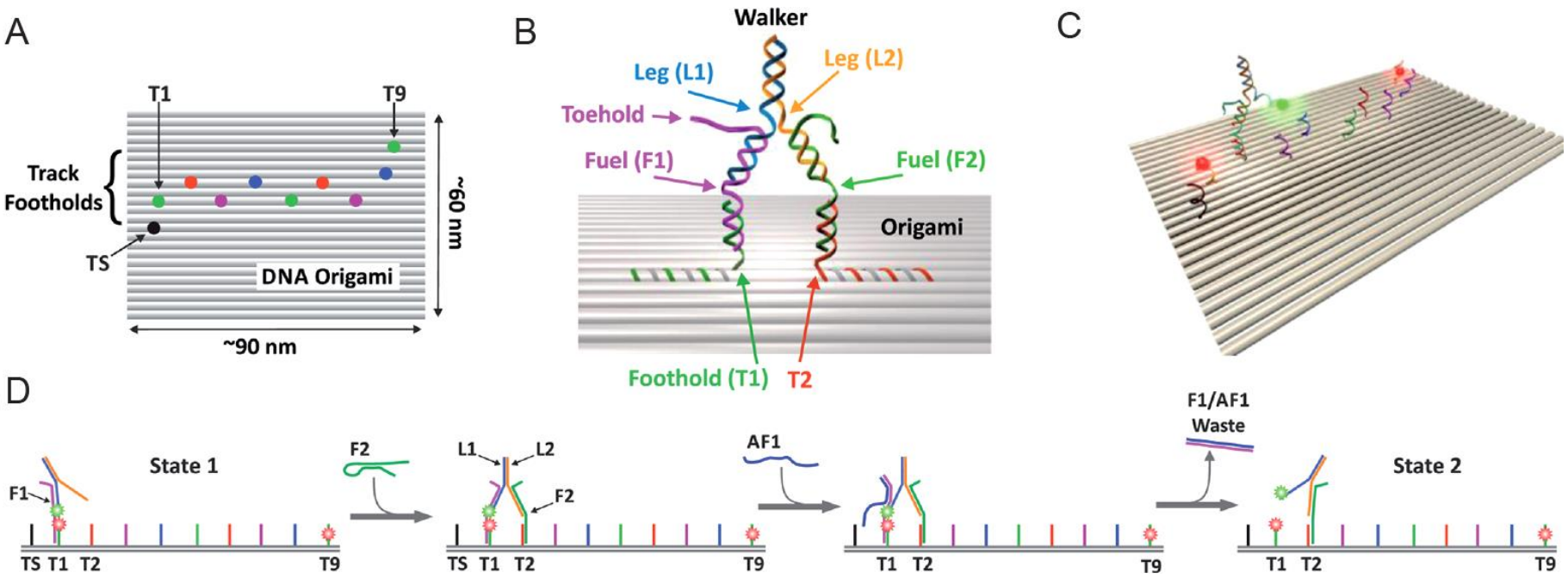
Actuating nanostructures with TMSD: tweezers



Yurke et al. Nature 406, 605–608, 2000

- DNA tweezers: First example of a DNA "nanomachine"
- Open and close drive by TMSD
- A fuel strand (F) closes the tweezers
- An anti-fuel strand (F*) removes F using TMSD
- Free energy for opening/closing comes from F-F* hybridization

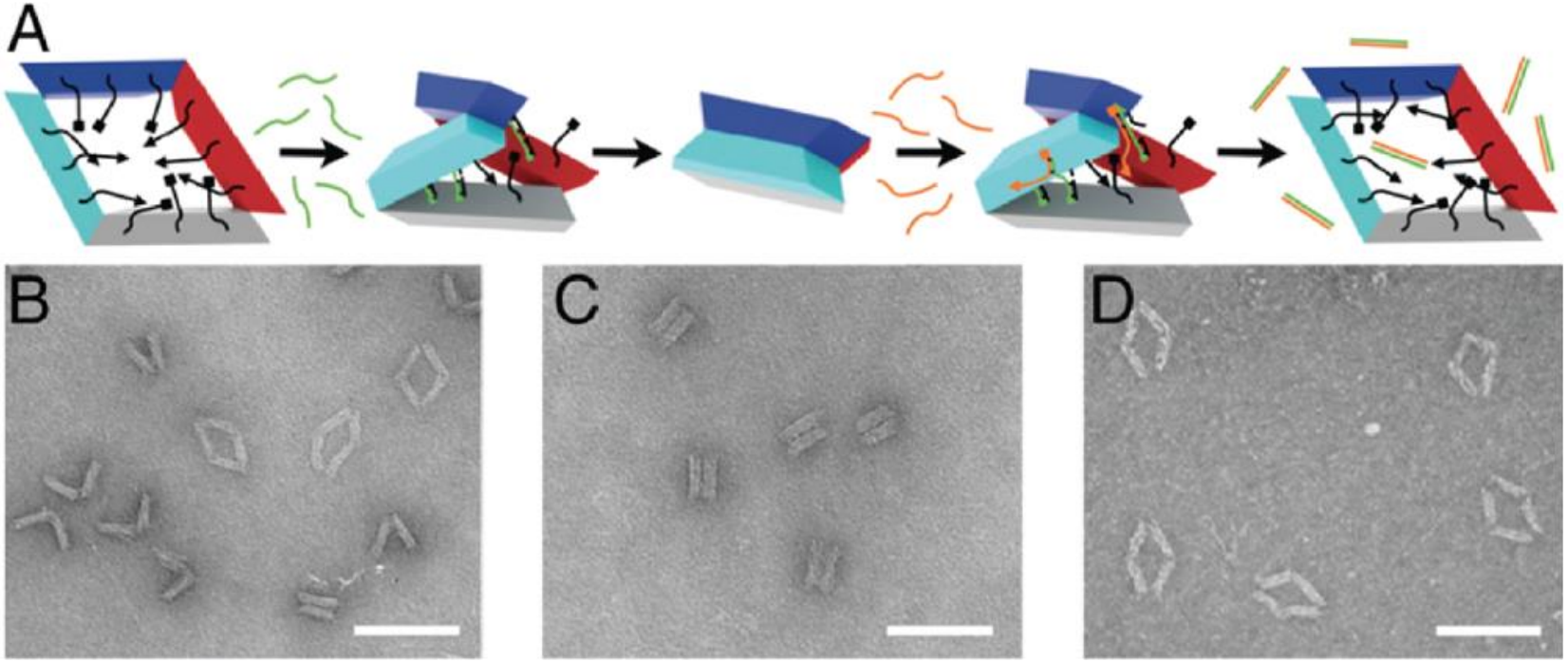
Actuating nanostructures with TMSD: walkers



DNA walker taking steps on a DNA origami. Binding and unbinding mediated by TMSD with fuel strands

Tomov et al. ACS Nano 11, 4002−4008, 2017

Actuating nanostructures with TMSD: origami



DNA origami in the shape of a "Bennet linkage" reconfigures with TMSD

Marras et al., PNAS, 112, 713−718, 2015